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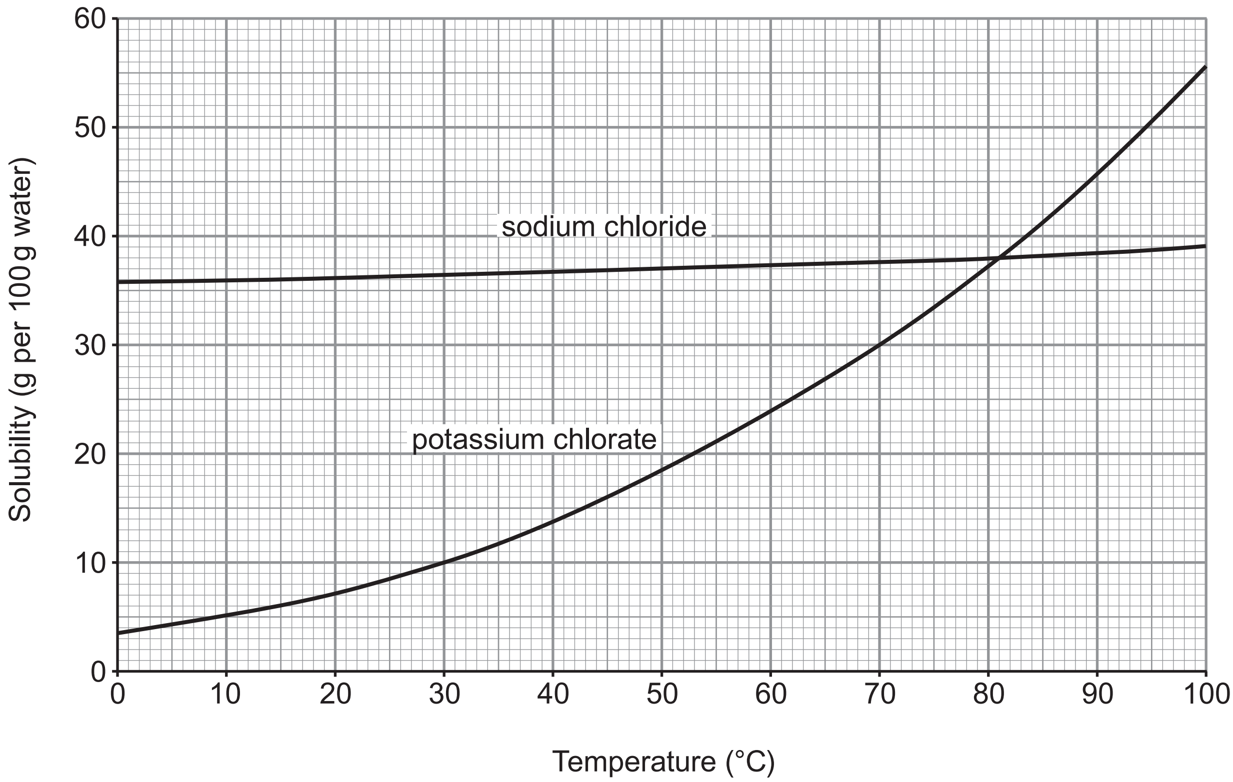
5. *(a)* The graphs below show the solubilities of sodium chloride and potassium chlorate in water at different temperatures.



(i) State the temperature at which the two compounds have the same solubility. [1]

.... °C

(ii) Calculate the mass of solid potassium chlorate that forms when a saturated solution in 100 g of water at 70 °C cools to 30 °C. [2]

mass = ... g

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(b) (i) Calculate the relative formula mass (M_r) of potassium chlorate, $KClO_3$. [2]

relative formula mass = ...

(ii) Calculate the percentage by mass of potassium in potassium chlorate. [2]

percentage by mass = ... %

(c) Potassium chlorate forms potassium chloride and oxygen on heating.

Balance the symbol equation that represents the reaction taking place. [1]

$$2KClO_3 \longrightarrow \boxed{}\ KCl + \boxed{}\ O_2$$

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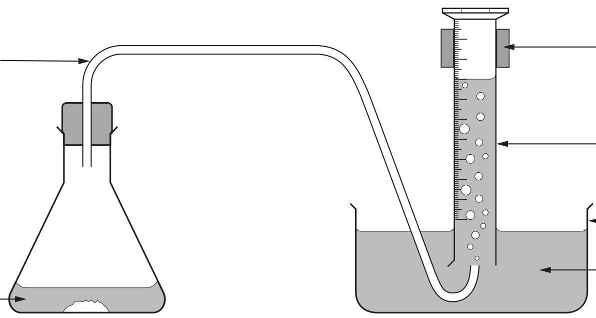
5. Hydrogen peroxide decomposes to give water and oxygen.

(a) Complete the symbol equation to show the reaction taking place. [2]

$$2H_2O_2 \longrightarrow \boxed{}\ H_2O + \ldots\ldots\ldots\ldots$$

(b) The rate of decomposition of hydrogen peroxide can be measured using the following apparatus.



The rate was investigated using three different catalysts. The results are shown in the table.

Time (s)	Volume of gas collected (cm^3)		
	Catalyst 1	Catalyst 2	Catalyst 3
0	0	0	0
20	2	20	8
40	4	34	15
60	6	38	23
80	8	40	30
100	10	40	36

(i) State which is the **least** effective catalyst. Give a reason for your answer. [1]

....

....

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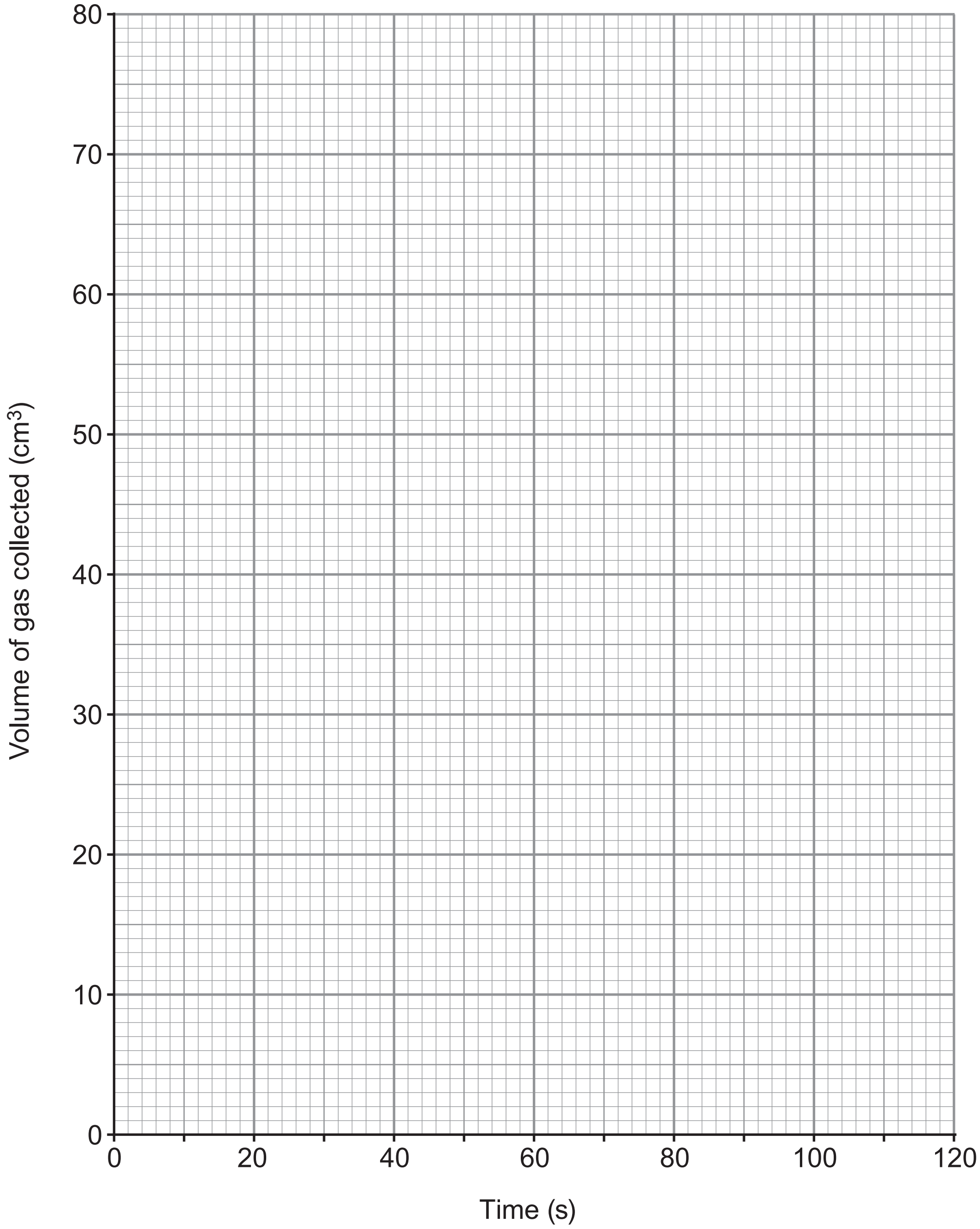
(ii) Plot a graph of the volume of gas collected using **catalyst 2**.
Draw a suitable line. [3]



(iii) **On the same grid**, sketch the graph you would expect to obtain if you added the same amount of catalyst 2 to the same volume of hydrogen peroxide of **twice** the concentration. [2]

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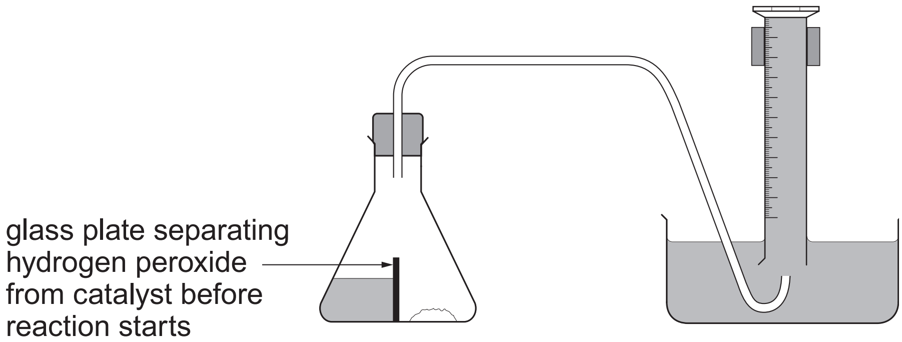
(iv) Another student claimed that he could collect more accurate results using the following apparatus.



Suggest how this apparatus could improve the accuracy of the results. [2]

....

....

....

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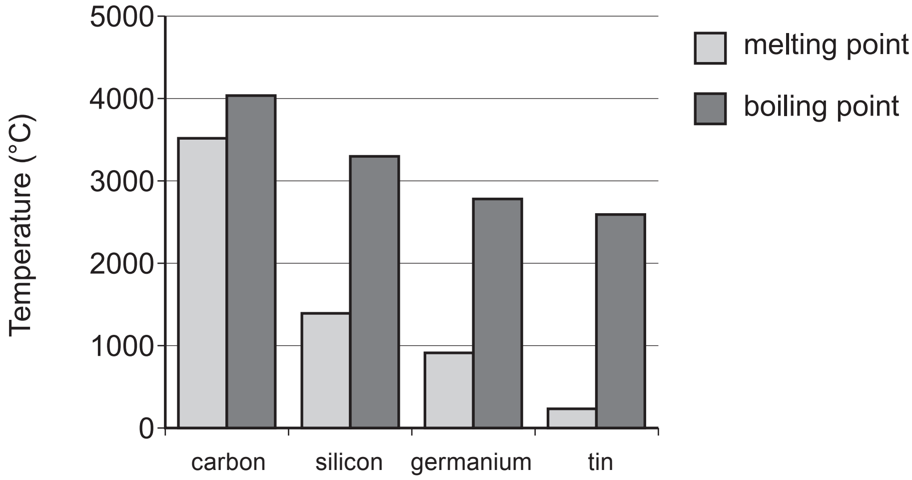
5. *(a)* The chart below shows the melting points and boiling points of some Group 4 elements.



(i) Describe the trend in melting point going down Group 4. [1]

....

(ii) Name the element which has the greatest **difference** between its melting point and boiling point. [1]

....

(b) (i) Name an element in Group 4 that is considered to be a semi-metal (metalloid). [1]

....

(ii) Tick (✓) the box below that describes where semi-metals are found in the Periodic Table. [1]

some are found in all groups	☐
they are all found in Group 4	☐
some are found in Group 1 and Group 2	☐
they are all found between metals and non-metals	☐

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(c) Carbon is the main element found in coal. When carbon burns it forms carbon dioxide.

(i) Write a **word** equation to show the reaction taking place. [1]

....

(ii) Name **one** environmental problem caused by increased levels of carbon dioxide in the atmosphere. [1]

....

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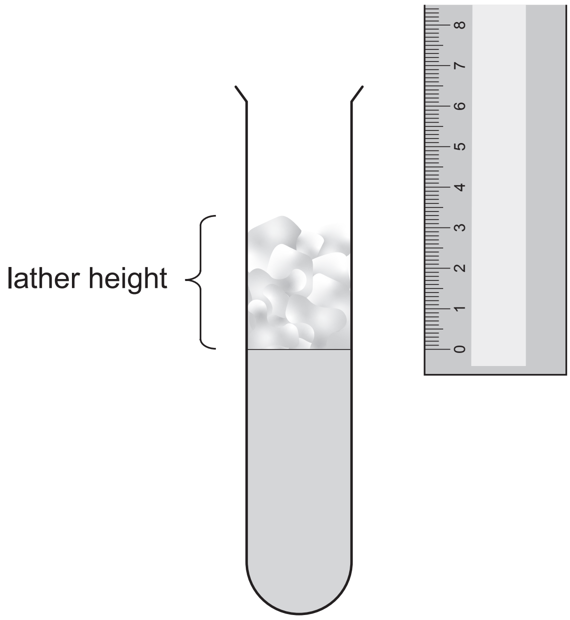
5. (a) A class measured the hardness of five different samples of water, **A**, **B**, **C**, **D** and **E**.

They added soap solution to each water sample and measured the height of the lather produced on shaking.



Their results are shown in the table.

Water sample	Lather height (cm)
A	1.5
B	3.0
C	0.5
D	5.0
E	2.0

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(i) Complete the order of hardness of the water samples, from softest to hardest. [1]

softest ⟶ hardest

D				**C**

(ii) Calcium is one of two metal ions which cause hardness in water.

Underline the name of the other metal ion which causes hardness in water. [1]

potassium **sodium** **nickel** **magnesium** **tin**

(iii) Tick (✓) **all** of the variables which must be controlled in this experiment. [2]

volume of soap solution	☐
type of water	☐
type of soap solution	☐
volume of water	☐
height of lather	☐
width of test tube	☐

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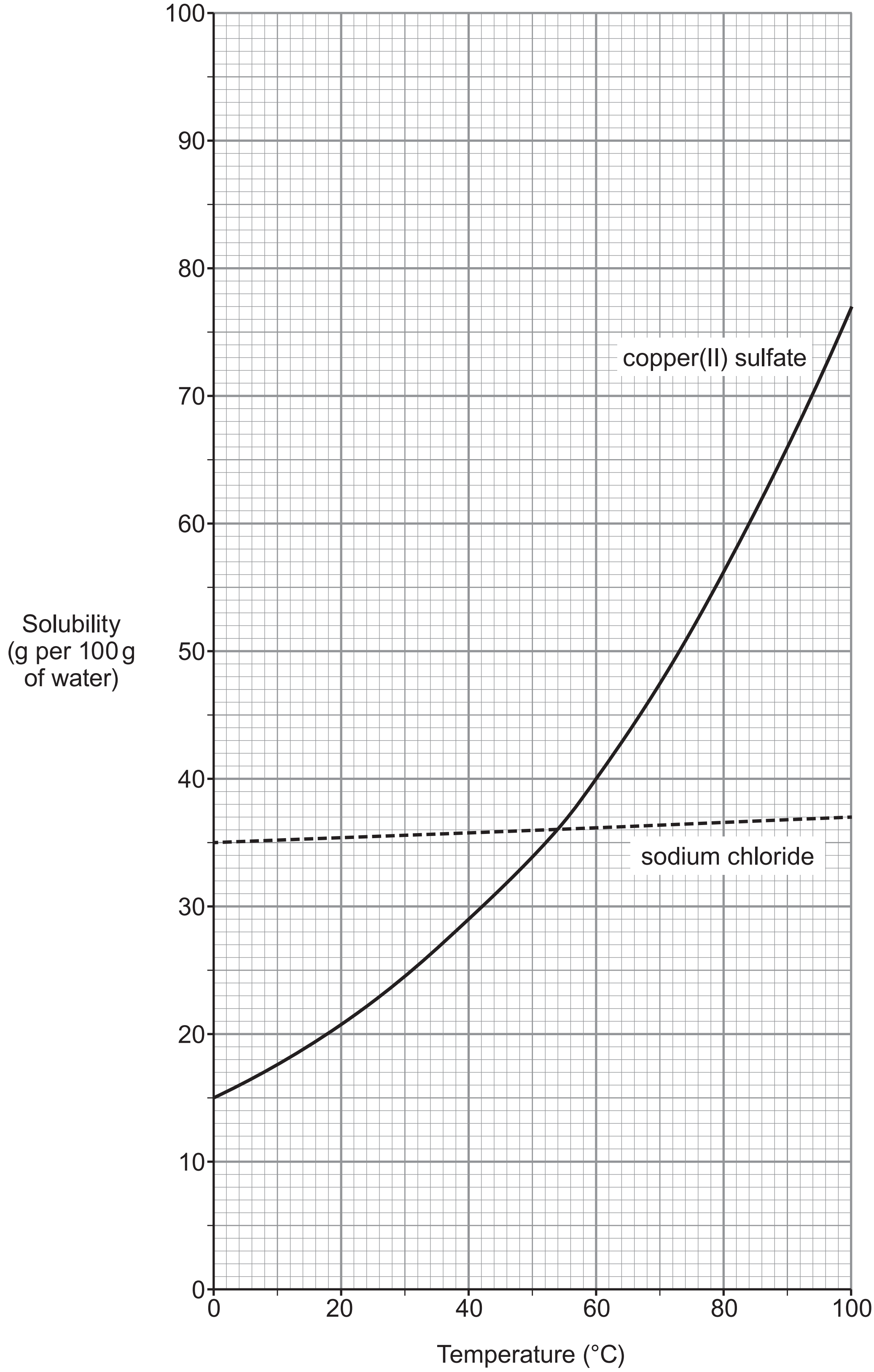
(b) The graph shows the solubilities of sodium chloride and copper(II) sulfate at different temperatures.



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(i) State the temperature at which the two compounds have the same solubility. [1]

.... °C

(ii) Compare the changes in solubility of the two compounds. [2]

....

....

....

(iii) Calculate the mass of copper(II) sulfate that would dissolve in **1000 g** of water at 40 °C. [2]

Mass = ... g

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(c) Potassium sulfate, K_2SO_4, is soluble in water.

(i) State the number of sulfur atoms in the formula K_2SO_4. [1]

....

(ii) Calculate the relative formula mass (M_r) of K_2SO_4. [2]

A_r(K) = 39 A_r(S) = 32 A_r(O) = 16

M_r = ...

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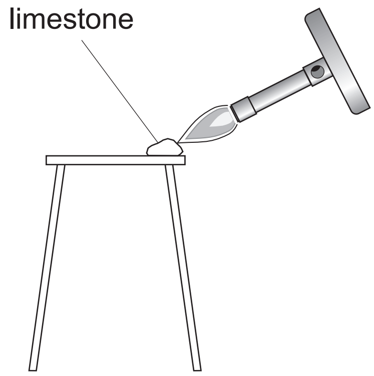
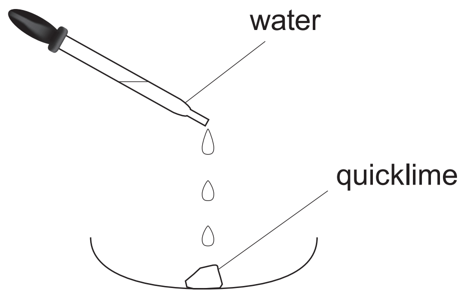
5. (a) A student carried out a two-stage experiment to change limestone (calcium carbonate) into slaked lime (calcium hydroxide).



Stage **1**: Limestone (calcium carbonate) decomposes into quicklime (calcium oxide) and carbon dioxide



Stage **2**: Quicklime (calcium oxide) reacts with water forming slaked lime (calcium hydroxide)

(i) Write the formulae for calcium oxide and carbon dioxide to complete the equation for the reaction taking place in stage **1**. [2]

$CaCO_3 \longrightarrow$ .... + ...

(ii) Calcium hydroxide contains one Ca^{2+} ion for every two OH^- ions.

Write the chemical formula for calcium hydroxide. [1]

....

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(b)

Describe the economic benefits and environmental drawbacks of limestone quarrying.
[6 QER]

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